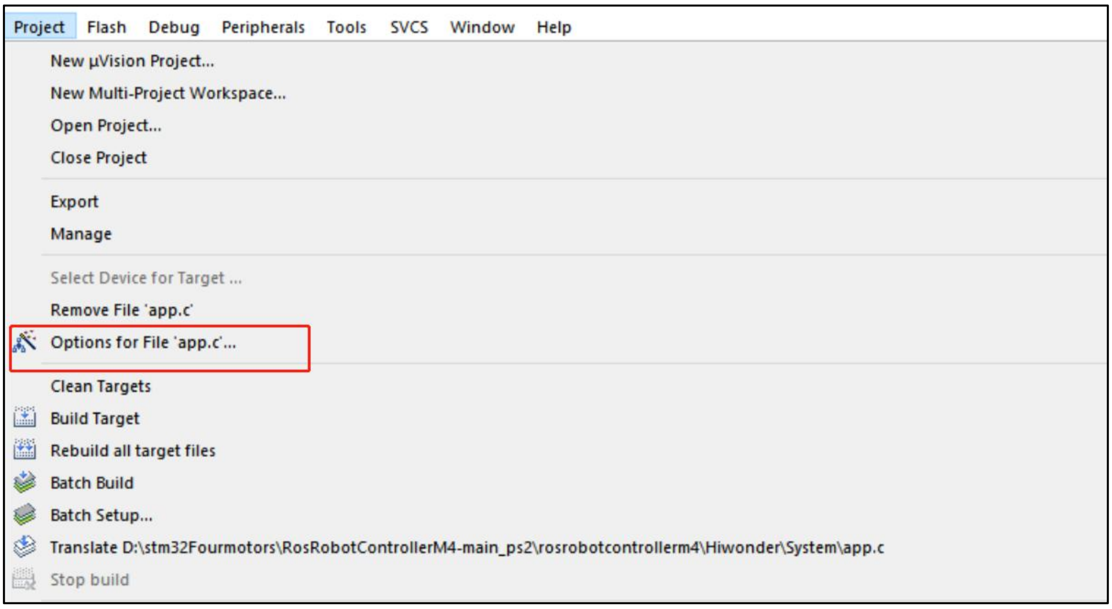
Project Compilation & Download

1. Project Compilation

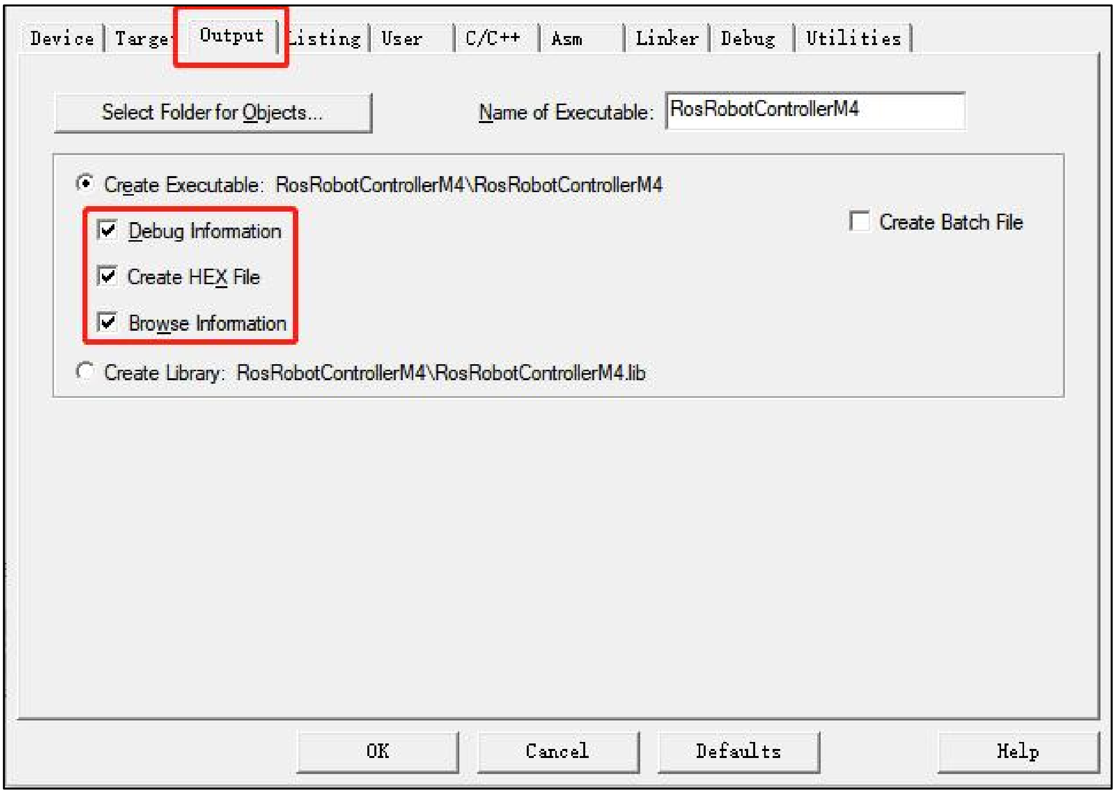
Once the program is written, we need to compile it into machine language for execution on the embedded system. Keil5 comes with a built-in compiler that can translate program source code into executable files. The compiler can generate different target files based on different processor architectures and instruction sets. Additionally, the compiler can optimize the code to produce more efficient and stable executable files. The process of project compilation involves the following steps:

1) Open the option to generate the hex file in settings:

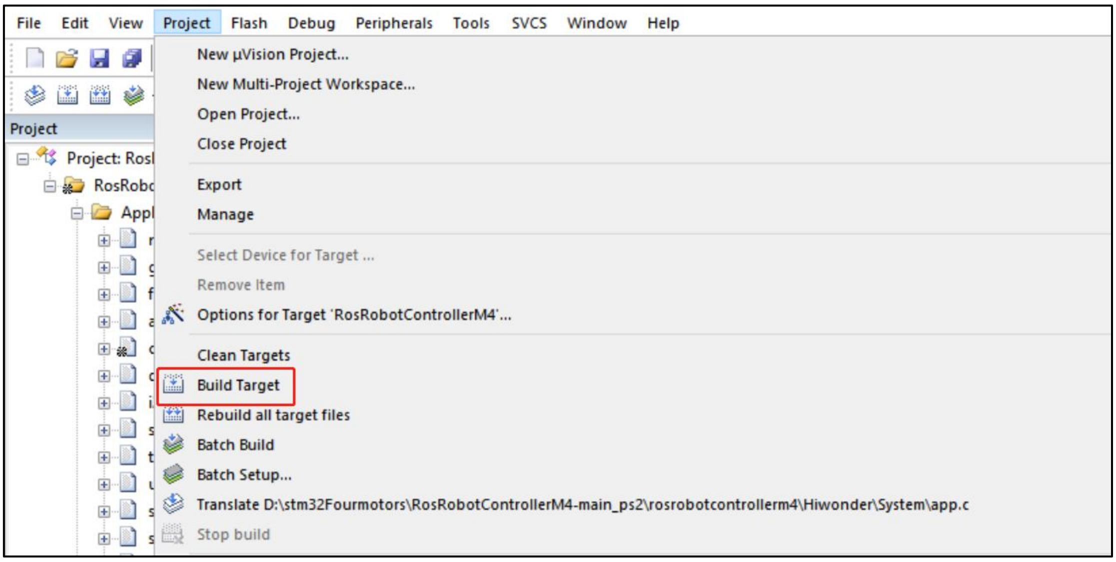
In the Keil5 software interface menu bar, click on the 'Project' option, then click on the 'Options for File 'app.c'' button.



Click on the "OutPut" option, tick the three red options below, and then click "OK".



2) In the Keil5 software interface menu bar, click on the "**Project**" option, then select "**Build Target**" from the drop-down menu that appears to compile the project, as shown in the figure below:



You can also compile by clicking on the corresponding icon on the toolbar, as shown in the figure below:

The following message appeared in the "**Build Output**" window at the bottom of the software, indicating a successful compilation.

```
Build Output
Build started: Project: RosRobotControllerM4
*** Using Compiler 'V5.06 update 6 (build 750)', folder: 'D:\Keil_v5\ARM\ARMCC\Bin'
Build target 'RosRobotControllerM4'
linking...
Program Size: Code=55036 RO-data=1916 RW-data=444 ZI-data=96340
FromELF: creating hex file...
"RosRobotControllerM4\LED.axf" - 0 Error(s), 0 Warning(s).
Build Time Elapsed:  00:00:03
```

Note: If the "Build Output" window displays error message(s) after compiling the project, it is necessary to double-click on the corresponding line to navigate to the relevant position and make corrections. Afterward, compile again. If there are warning message(s), these can be ignored.

2. USB Download

After compiling the project, you can download the generated hex file to the STM32 main control board. The following software and hardware materials are required.

2.1 Software & Hardware Preparation

Software: mcuisp (included in the attachments). For specific installation instructions, please refer to "Appendix/Software Installation".

Hardware: Type-C cable and STM32 controller

2.2 Download Steps

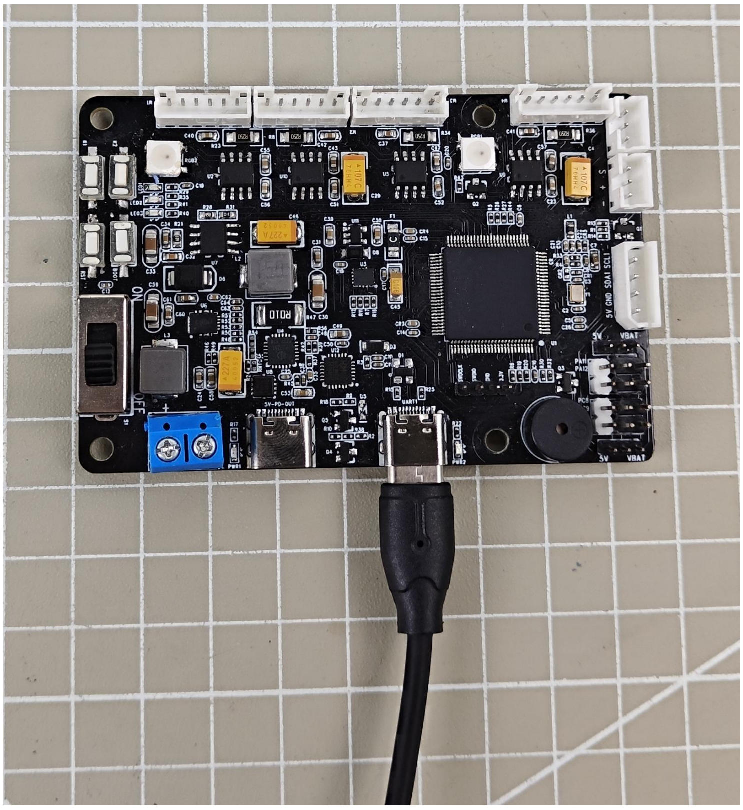
1. Type-C to USB cable is used to connect the computer and the STM32 main control board (hereinafter referred to as Type-C cable).

2. STM32 Controller

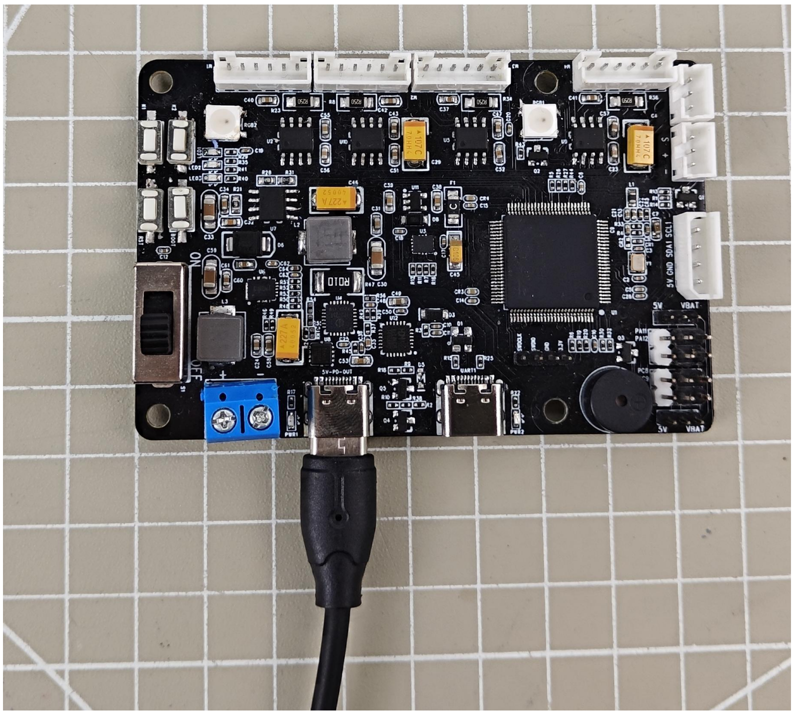
The specific operation steps are as follows, with the programmed named "Blinking LED":

（1）**Hardware Connection:**

Insert the Type-C cable into the Type-C port (UART1) of the STM32 main control board and connect it to the USB port of the computer.



[UART1]

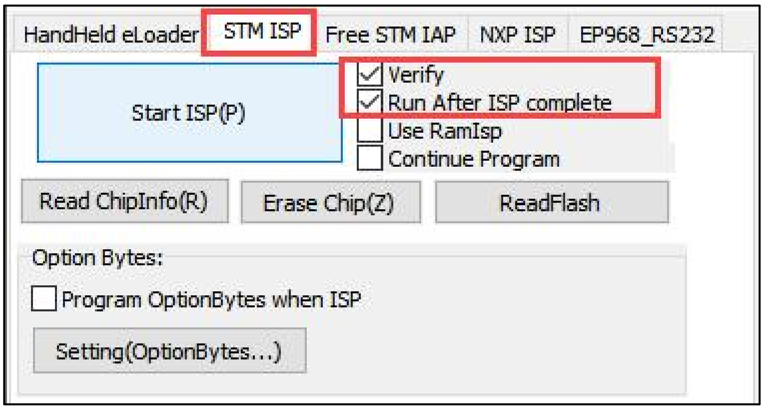


[power supply]

（2）Basic Configuration:

Open the mcuisp software, click on "**Search Serial Port**" in the menu bar at the top of the software, then set the baud rate (bps) to 115200.

Click on the "STMISP" option in the software interface.

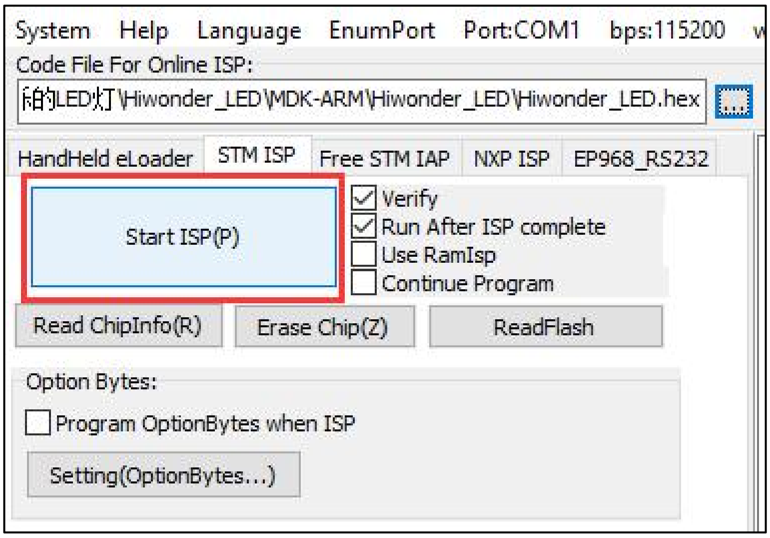


Select "DTR low level reset, RTS high level to enter BootLoader" at the bottom.

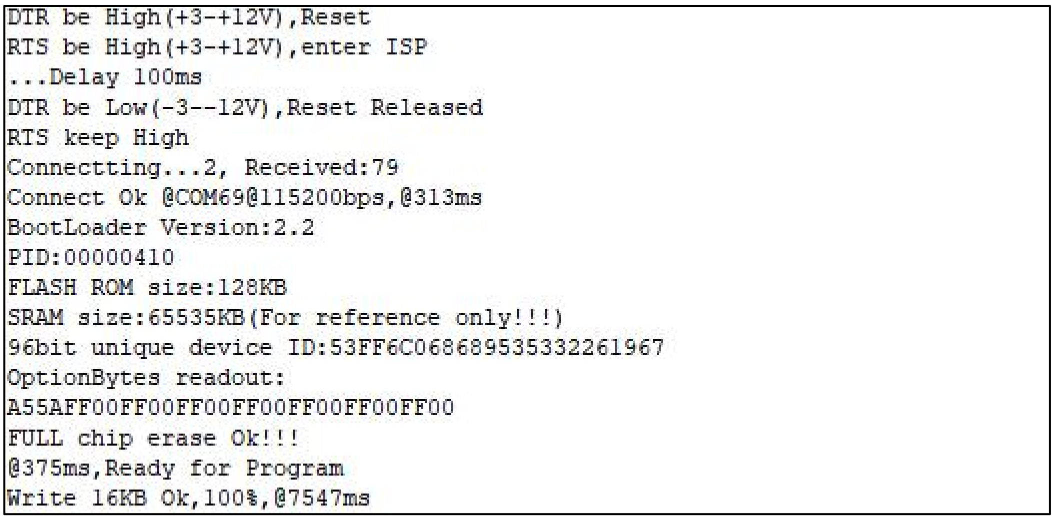
（3）Software Burning:

Click on the button inside the red box in the mcuisp software interface, and select the hex file that needs to be burned.

Return to the previous interface and click on the "Start ISP" button to burn the generated hex file onto the STM32 main control board.



Burning in progress



If the sidebar displays the following image, it indicates that the burning process is complete.

Note: The STM32 main control board comes with pre-loaded software (resulting in LED1 blinking blue every 1 second). If you need to burn the PS2 joystick control program, you can find the RosRobotControllerM4.hex file in the 'Chapter 1: Introduction to the Car Hardware and Usage / Section 2: PS2 Joystick Remote Control / Program / rosrobotcontrollerm4_motor_ps2 / MDK-ARM / RosRobotControllerM4' folder for burning.

```
DTR be High(+3-+12V),Reset
RTS be High(+3-+12V),enter ISP
...Delay 100ms
DTR be Low(-3--12V),Reset Released
RTS keep High
Connectting...2, Received:79
Connect Ok @COM69@115200bps,@313ms
BootLoader Version:2.2
PID:00000410
FLASH ROM size:128KB
SRAM size:65535KB(For reference only!!!)
96bit unique device ID:53FF6C06868953533226l967
OptionBytes readout:
A55AFF00FF00FF00FF00FF00FF00FF00
FULL chip erase Ok!!!
@375ms,Ready for Program
Write 16KB Ok,100%,@7547ms
www.mcuisp.com:Mission Complete,Anything Ok!!!
```

If burning fails, refer to the 'Troubleshooting Guide' document in the 'Appendix' folder for assistance.

Note: To avoid any abnormal occurrences during the burning process, users should follow the steps in sequence carefully!!!
